

where r_t^{R1000} represents the return of the Russell 1000 and ε_t is the residual of the regression.

The $\alpha = 3.44\%$ per year in this CAPM regression is the average excess return of the low volatility strategy relative to a *market-adjusted portfolio*. Put another way, we can rewrite the CAPM regression using a benchmark portfolio of a risk-free asset and 0.73 of the Russell 1000:

$$r_t = 0.0344 + \underbrace{0.2728r_t^f + 0.7272r_t^{R1000}}_{r_t^{bmk}} + \varepsilon_t,$$

That is, suppose the benchmark $r_t^{bmk} = 0.27r_t^f + 0.73r_t^{R1000}$ consists of a portfolio holding 27% in the risk-free asset and 73% in the Russell 1000. The low volatility strategy outperforms this benchmark by 3.44% per year. The information ratio of the low volatility strategy with this *risk-adjusted benchmark* is a very high 0.78.

If we assume a naive benchmark of just the Russell 1000, we falsely assume that the beta of the low volatility strategy is one (when in fact it is 0.73). With the Russell 1000 benchmark we have

$$r_t = 0.0150 + \underbrace{r_t^{R1000}}_{r_t^{bmk}} + \varepsilon_t,$$

so the alpha is 1.50% per year. The information ratio of the low volatility strategy relative to the naive Russell 1000 benchmark is just 0.24. This is not the correct risk-adjusted benchmark because the beta of the low volatility strategy is not one.

Thus, even with a simple Russell 1000 portfolio, failing to adjust the benchmark for risk can make a huge difference in the alpha!

Ideal Benchmarks

What are the characteristics of a sound benchmark? It should be:

1. Well defined

Produced by an independent index provider, the Russell 1000 is verifiable and free of ambiguity about its contents. Thus it ably defines the “market portfolio” for GM and Martingale.

2. Tradeable

Alpha must be measured relative to tradeable benchmarks, otherwise the computed alphas do not represent implementable returns on investment strategies.

So the benchmark should be a realistic, low-cost alternative for the asset owner. The Russell 1000 is a natural passive benchmark for Martingale’s low volatility strategy because low-cost mutual fund and ETF versions are available (see chapter 16).